

gemstone-py Companion Manual

Setup guide, user manual, examples guide, and cookbook in one volume

Generated from the Markdown sources in `gemstone-py/docs`

Assets are local SVG illustrations stored in the repository.

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gemstone-py Documentation

This directory contains the longer-form documentation set for `gemstone-py`. The top-level `README.md` in the repo is still the quickest way to get the package installed, but the files here are meant to answer the questions people ask on day two, day ten, and day forty-five.

Reading Order

If you are new to the project, use this order:

1. [Setup Guide](#)
2. [User Manual](#)
3. [Examples Guide](#)
4. [Cookbook](#)
5. [Funny Introduction](#)

If you want a narrative overview before diving in:

- [Medium article](#) — a complete end-to-end guide written in article style

If you are already productive and only need answers:

- "Why is my login failing?" → [Setup Guide](#)
- "Which transaction policy should I use?" → [User Manual](#)
- "Which example should I run first?" → [Examples Guide](#)
- "How do I do X quickly?" → [Cookbook](#)
- "I want the whole story, plus jokes." → [Funny Introduction](#)

What This Docs Set Covers

- installing and configuring `gemstone-py`
- connecting to a GemStone stone from Python
- transaction policies, session scopes, and failure behaviour
- async sessions and FastAPI request integration
- typed OOPs, typed `GSCollection` queries, and managed OOP lifetimes
- optional native PyO3 backend installation and backend selection
- the persistent data helpers:
 - `PersistentRoot`
 - `GSCollection`
 - `GStore`
 - `ObjectLog`
- concurrency helpers such as `RCCounter`, `RCHash`, and `RCQueue`

- Flask request-session integration
- benchmarks, release workflows, and the current examples directory

Visuals

The images under `docs/assets/` are intentionally repository-native SVG files. That gives you a few benefits:

- they render nicely on GitHub
- they can be edited in a normal text diff
- they do not bloat the repository with binary noise
- they can be reused in presentations, blog posts, or generated manuals

Some of the "screenshots" are stylized screenshot illustrations rather than raw captures. That is deliberate: the examples evolve, and the docs should remain easy to maintain.

Print-Friendly Book

The long-form introduction under `funny-introduction/` is structured as a book, with explicit page breaks for a print/export workflow. It is designed to compile to more than one hundred pages when rendered to PDF or another paged format.

Suggested Build / Export Flow

If you want to turn these docs into a PDF bundle later, a practical approach is:

1. keep the Markdown source here as the canonical text
2. render the long introduction as a separate book
3. render the setup guide, manual, examples guide, and cookbook as a smaller companion manual

That split keeps the funny book delightfully excessive and the operational docs pleasantly searchable.

Setup Guide

This guide gets you from "I cloned the repo" to "I can log in to GemStone from Python and run the examples without muttering at the terminal."

What You Need

At minimum:

- Python 3.11 or newer
- access to a GemStone/S 64 stone
- the GemStone client library (`libgcirpc`)
- a username and password that can log into the stone

For the full local development experience, you will also want:

- the repository checkout
- a working virtual environment
- the ability to run live tests against your stone
- if you use the self-hosted workflows, a GitHub Actions runner on the GemStone host

Install the Package

From PyPI:

```
python3 -m pip install gemstone-py
```

If you want the optional PyO3 native fast path and a wheel exists for your platform:

```
python3 -m pip install "gemstone-py[fast]"
python -c "from gemstone_py import _gci; print(_gci.IMPLEMENTATION)"
```

From a repository checkout, the fuller backend example is `python -m examples.native_backend.check_backend`.

From a repo checkout:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -U pip
python -m pip install -e .[dev]
```

Required Environment Variables

The package expects its runtime configuration from environment variables. The minimum set for most local work is:

```
export GS_LIB=/opt/gemstone/product/lib
export GS_STONE=gs64stone
export GS_USERNAME=DataCurator
export GS_PASSWORD=swordfish
```

Common optional variables:

```
export GS_HOST=localhost
export GS_NETLDI=netldi
export GS_GEM_SERVICE=gemnetobject
export GS_HOST_USERNAME=
export GS_HOST_PASSWORD=
export GS_LIB_PATH=/full/path/to/libgcirpc-3.7.x-64.dylib
```

Quick reference:

Variable	Required	Example	Purpose
GS_LIB	Yes	/opt/gemstone/product/lib	GemStone lib/ directory for library discovery
GS_STONE	Yes	gs64stone	Stone name
GS_USERNAME	Yes	DataCurator	GemStone login username
GS_PASSWORD	Yes	swordfish	GemStone login password
GS_LIB_PATH	No	/full/path/libgcirpc.dylib	Pin to a specific libgcirpc file
GS_HOST	No	localhost	Remote stone host
GS_NETLDI	No	netldi	NetLDI service name
GS_GEM_SERVICE	No	gemnetobject	Gem service name
GS_HOST_USERNAME	No		Remote host OS username
GS_HOST_PASSWORD	No		Remote host OS password

First Real Login

The simplest sanity check is a tiny Python script:

```
from gemstone_py import GemStoneConfig, GemStoneSession

config = GemStoneConfig.from_env()

with GemStoneSession(config=config) as session:
    print(session.eval("1 + 2"))
```

If that prints `3`, the package can:

- read your configuration
- load `libgcirpc`
- log in to GemStone
- evaluate Smalltalk in the repository

That is enough to start.

Recommended First Commands

The installed package ships with a few small CLI helpers:

```
gemstone-hello
gemstone-smalltalk-demo
gemstone-examples hello
gemstone-examples smalltalk-demo
gemstone-benchmarks --help
```

From a repository checkout, these examples are also useful first checks:

```
python -m examples.native_backend.check_backend
```

What they are good for:

- `gemstone-hello`

Tiny smoke check for the installed package.

- `gemstone-smalltalk-demo`

Basic bridge demo that is easy to reason about.

- `gemstone-examples ...`

A stable wrapper for the example entry points.

- `gemstone-benchmarks`

The maintained benchmark lane, distinct from the teaching examples.

- `examples.native_backend.check_backend`

Shows whether the current process selected ctypes or the optional native GCI backend.

Transaction Policy: Read This Early

The most important behavioural rule in `gemstone-py` is that transaction intent should be explicit.

`GemStoneSession(...)` defaults to manual transaction control.

That means:

- a plain session does not silently commit just because your `with` block ended
- write scripts must call `commit()` explicitly or use a scoped helper that commits on success
- read-only scripts should usually abort or use an abort-on-exit policy

The common options are:

```
from gemstone_py import GemStoneSession, TransactionPolicy

with GemStoneSession(
    config=config,
    transaction_policy=TransactionPolicy.COMMIT_ON_SUCCESS,
) as session:
    ...
```

Or:

```
from gemstone_py import session_scope

with session_scope(config=config) as session:
    ...
```

Use this rule of thumb:

- one-off write script -> `COMMIT_ON_SUCCESS`
- read-only inspection -> `ABORT_ON_EXIT`
- library or framework code -> manual, unless you deliberately wrap it

First Useful Persistent Write

Once login works, prove that you can store and read data:

```
from gemstone_py import GemStoneConfig, TransactionPolicy
from gemstone_py import GemStoneSession
from gemstone_py.persistent_root import PersistentRoot
```

```

config = GemStoneConfig.from_env()

with GemStoneSession(
    config=config,
    transaction_policy=TransactionPolicy.COMMIT_ON_SUCCESS,
) as session:
    root = PersistentRoot(session)
    root["DocsSmokeTest"] = {"status": "ok", "kind": "setup-guide"}

with GemStoneSession(config=config) as session:
    root = PersistentRoot(session)
    print(root["DocsSmokeTest"])

```

If that round trip works, the rest of the package will feel much less mysterious.

First Async Check

If your application is async-first, run the async example after the basic sync login works:

```
python -m examples.async_features.session_root_and_collection
```

The example opens an `AsyncSession`, writes through `AsyncPersistentRoot`, keeps a managed OOP alive, and exercises `AsyncGSCollection`.

For FastAPI:

```
python -m pip install fastapi uvicorn
uvicorn examples.fastapi.app:app --reload
```

Visit <http://127.0.0.1:8000/health/gemstone>.

Typed and Lifetime Examples

Once basic reads and writes work, these examples show the newer type and export-set APIs:

```
python -m examples.typed_access.typed_oops_and_queries
python -m examples.lifetime.managed_oop_handles
```

Use them to confirm that `TypedOop[T]`, typed `GSCollection` predicates, `execute_managed(...)`, and `session.handle(...)` behave against your stone.

Running the Tests

Unit tests:

```
python -m unittest discover -s tests -p 'test*.py'
```

The standard local verification lane:

```
./scripts/run_ci_checks.sh
```

Live GemStone tests:

```
GS_RUN_LIVE=1 ./scripts/run_live_checks.sh
```

Longer live soak tests:

```
GS_RUN_LIVE=1 GS_RUN_LIVE_SOAK=1 ./scripts/run_live_checks.sh
```

Destructive live tests are intentionally separated. They mutate shared state and are not meant to be run casually.

Troubleshooting

GemStoneConfigurationError

Usually means `GS_USERNAME` and/or `GS_PASSWORD` are missing.

OSError while loading the client library

Usually one of:

- `GS_LIB` is wrong
- `GS_LIB_PATH` points at a missing or incompatible `libgcirpc`
- the local machine does not actually have the client library installed

Login works in one shell but not another

Check that both shells export the same environment values. This failure mode is common, boring, and surprisingly effective.

Your writes "worked" but the data is missing

That almost always means you forgot to commit or assumed a session would commit for you. Re-check the transaction policy.

A Flask request wrote partial data after a handled error

Use the request-session integration from `gemstone_py.web` and let request teardown own the final commit-or-abort decision. The web helpers were hardened exactly to avoid this problem.

The native backend is not selected

Check the current process:

```
python -m examples.native_backend.check_backend
```

If `selected backend` is `ctypes`, either the optional package is not installed or `GEMSTONE_PY_GCI_BACKEND=ctypes` was set before Python started. Install `gemstone-py[fast]` and start a fresh Python process. Use `GEMSTONE_PY_GCI_BACKEND=native` when you want import failure instead of silent fallback.

Where to Go Next

- [User Manual](#) for the main abstractions
- [Examples Guide](#) for the runnable demos
- [Cookbook](#) for copy-paste-friendly recipes

User Manual

This manual explains the public surface of `gemstone-py` from the point of view of a user who wants to build reliable code, not just make a demo print a nice dictionary once.

The Mental Model

Three ideas matter more than the rest:

1. Python code drives the application.
2. GemStone stores the persistent state.
3. Transactions are explicit enough that you can explain them to another human.

`gemstone-py` is not trying to hide GemStone completely. It gives you a Pythonic surface, but it still wants you to understand:

- when sessions are opened and closed
- when transactions commit or abort
- which data structures are persisted in GemStone
- what happens when two sessions try to change the same thing

That restraint is a strength. The library does not pretend that distributed state is a teddy bear.

Core Building Blocks

GemStoneConfig

`GemStoneConfig` is the runtime configuration object. In most applications you will construct it from the environment:

```
from gemstone_py import GemStoneConfig

config = GemStoneConfig.from_env()
```

Use it when:

- you want a single source of connection settings
- you need to pass the same config into sessions, stores, or web providers
- you want to validate missing credentials early

GemStoneSession

This is the direct session object. Use it when you want explicit session and transaction control.

Typical pattern:

```
from gemstone_py import GemStoneSession, TransactionPolicy

with GemStoneSession(
    config=config,
    transaction_policy=TransactionPolicy.COMMIT_ON_SUCCESS,
) as session:
    print(session.eval("System myUserProfile userId"))
```

Useful methods include:

- `eval(...)`

Evaluate Smalltalk code directly.

- `commit()`

Commit the current transaction.

- `abort()`

Abort the current transaction.

- `global_get(...)`

Resolve named objects from the repository.

When to use it:

- scripts
- lower-level tooling
- integration code that must control failure behaviour precisely

`session_scope(...)`

This is the higher-level unit-of-work helper. It is usually the right answer for application code.

```
from gemstone_py import session_scope

with session_scope(config=config) as session:
    ...
```

Why it exists:

- it makes commit-on-success intent obvious
- it composes naturally with service-layer code
- it avoids sprinkling `commit()` everywhere like nervous confetti

`AsyncSession`

`gemstone_py. aio.AsyncSession` is the async facade for applications built on `asyncio`, FastAPI, or Starlette-style request handling.

It does not make GCI itself async. Instead, it runs all calls for one session on one dedicated worker thread. That keeps the GemStone session on its owning thread while letting your event loop continue serving other work.

```
from gemstone_py import GemStoneConfig
from gemstone_py. aio import AsyncSession

config = GemStoneConfig.from_env()

async with AsyncSession.connect(config=config) as session:
    value = await session.eval("3 + 4")
```

```

async with session.transaction():
    root = session.root()
    await root.set("AsyncManualExample", {"value": value})

```

Use `AsyncSession` when:

- request handlers are already async
- you want FastAPI dependency injection
- you need to keep blocking GCI work out of the event loop

The runnable repository example is `examples/async_features/session_root_and_collection.py`.

Transaction Policies

`TransactionPolicy` exists so you do not have to guess what a context manager will do.

The main values are:

- `MANUAL`
- `COMMIT_ON_SUCCESS`
- `ABORT_ON_EXIT`

Recommended use:

Situation	Policy
low-level explicit control	<code>MANUAL</code>
script that writes data	<code>COMMIT_ON_SUCCESS</code>
read-only inspection	<code>ABORT_ON_EXIT</code>
request/response web unit of work	<code>COMMIT_ON_SUCCESS</code> via <code>session_scope(...)</code> or request integration

PersistentRoot

`PersistentRoot` is the friendliest entry point into repository-backed state.

The default instance is the user's `UserGlobals` dictionary:

```

from gemstone_py.persistent_root import PersistentRoot

```

```
root = PersistentRoot(session)
root["CustomerCount"] = 12
```

Other dictionaries are also available:

- `PersistentRoot.globals(session)`
- `PersistentRoot.published(session)`
- `PersistentRoot.session_methods(session)`

Use `PersistentRoot` when:

- you want a stable named entry point
- you are storing dictionaries, counters, queues, or domain objects
- you want something simple and direct

Use something more specialized when:

- you need indexed search across many rows -> `GSCollection`
- you want a file-like key/value store abstraction -> `GStore`
- you want append-only event style logging -> `ObjectLog`

Good Practices With `PersistentRoot`

- keep the top-level key space tidy
- use descriptive names
- group related data under one top-level bucket when it makes sense
- do not turn `UserGlobals` into a junk drawer with no naming discipline

GSCollection

`GSCollection` is the indexed collection helper. It is the right tool when you need more than "look up a dictionary by a single well-known key."

Use it for:

- indexed search
- filtering by stored attributes
- application collections with repeatable lookup patterns

Typical pattern:

```
from gemstone_py.gsquery import GSCollection

people = GSCollection("People", config=config)
people.insert({"@name": "Tariq", "@city": "London"})
```

```
people.add_index("@city")
matches = people.search("@city", "eq1", "London")
```

Think of it as the package's middle ground between:

- raw repository objects
- and a full ORM that wants to restructure your life

Typed Queries

`GSCollection.query(...)` can accept a Python `Protocol` as a type witness. The query still runs against GemStone indexed paths, but your Python code gets attribute names and better editor help.

```
from typing import Protocol
from gemstone_py.gsquery import GSCollection

class BlogPostRecord(Protocol):
    title: str
    status: str

posts = GSCollection("SimplePosts", config=config).query(BlogPostRecord)
published = posts.where(lambda post: post.status == "published").all()

for post in published:
    print(post.title)
```

The lambda is a query builder expression, not a live object callback. `post.status` becomes the GemStone ivar path `@status`.

See `examples/typed_access/typed_oops_and_queries.py` for a runnable typed query and `examples/typed_access/simple_blog_queries.py` for importable helper functions.

Typed OOPs and Lifetime Handles

Raw integer OOPs remain supported for compatibility. New code can opt into typed or managed wrappers when the extra structure is useful.

`TypedOop[T]` carries a phantom Python type:

```
from typing import Protocol
from gemstone_py import GemStoneSession, gemstone_class

@gemstone_class("Date")
class GemStoneDate(Protocol):
    @property
    def printString(self) -> str: ...
```

```
with GemStoneSession(config=config) as session:
    today = session.execute_typed("Date today", GemStoneDate)
    print(today.proxy().printString)
```

For object lifetime, use `execute_managed(...)` when a raw OOP should stay in GemStone's export set while Python holds a handle:

```
with GemStoneSession(config=config) as session:
    collection = session.execute_managed("OrderedCollection new")
    collection.send("add:", session.new_string("kept alive"))
    print(collection.print_string())
    collection.close()
```

Use `session.handle(oop)` when you already have a raw OOP and want an explicit scope:

```
with session.handle(raw_oop) as handle:
    print(handle.send("printString"))
```

`execute()` and `perform()` still return raw OOP integers. The managed variants are additive and make lifetime intent visible.

The runnable repository example is `examples/lifetime/managed_oop_handles.py`.

GStore

GStore is a GemStone-backed key/value store with its own transactional behaviour around store operations.

Use it when:

- you want a store-like API
- you want separate named stores
- you want a compact way to persist structured payloads

Typical pattern:

```
from gemstone_py.gstore import GStore

store = GStore("inventory.db", config=config)
store["sku:123"] = {"name": "Hat", "stock": 8}
print(store["sku:123"])
```

GStore is especially handy in examples, utilities, and small application components that want a store-shaped abstraction without building a full domain model first.

ObjectLog

`ObjectLog` is the package's event-log helper. It is good for:

- audit trails
- structured append-only logging
- "what happened?" questions after a workflow ran

Typical pattern:

```
from gemstone_py.objectlog import ObjectLog

log = ObjectLog(config=config)
log.info("user_created", {"user": "tariq", "source": "manual"})
for entry in log.entries():
    print(entry)
```

This is not a replacement for Python logging. It is a repository-resident event log for when the record should live with the data.

Concurrency Helpers

The concurrency helpers give you shared, repository-backed coordination primitives:

- `RCCounter`
- `RCHash`
- `RCQueue`

These are useful when you need shared mutable state between sessions and want the concurrency semantics to live in GemStone instead of in a Python process.

Use them carefully:

- they are powerful
- they make contention visible
- they will teach you honesty about multi-session behaviour

The package also includes:

- nested transaction helpers
- conflict detection helpers
- instance listing utilities

Commit Conflicts

When two sessions modify the same object and both try to commit, GemStone raises a conflict. The second committer gets a `CommitConflictError`.

This is not a bug. It is the correct behaviour of an optimistic concurrency system. The right response is:

1. catch `CommitConflictError`
2. call `session.abort()` to reset the transaction
3. reload the data
4. reapply the change (with a narrowed scope if possible)
5. retry the commit

Keep retries bounded. Log them. If you see frequent conflicts on the same object, that is a signal the write pattern needs rethinking — not that the concurrency model is wrong.

```
from gemstone_py import GemStoneSession, TransactionPolicy
from gemstone_py.concurrency import CommitConflictError

for attempt in range(3):
    with GemStoneSession(config=config,
                        transaction_policy=TransactionPolicy.MANUAL) as session:
        try:
            counter = session.global_get("Visits")
            session.eval(f"Visits := {counter + 1}")
            session.commit()
            break
        except CommitConflictError:
            session.abort()
            if attempt == 2:
                raise
```

Web Integration

The web integration lives in `gemstone_py.web`.

The main pieces are:

- `install_flask_request_session(...)`
- `GemStoneSessionPool`
- `GemStoneThreadLocalSessionProvider`
- `session_scope(...)`

One Session Per Request

If you are building a Flask app, use the request integration so the request lifecycle decides the final transaction outcome.

That avoids the worst class of web persistence bugs:

- request handled an exception
- response still rendered

- persistence layer silently committed partial work anyway

The package already corrected this behaviour at the framework layer, so use it.

Pool vs Thread-Local

Use `GemStoneSessionPool` when:

- you have multiple request workers
- you want bounded reuse
- you want production-friendly pooling semantics

Use `GemStoneThreadLocalSessionProvider` when:

- your threading model is simple and stable
- one session per thread is the clearest fit

FastAPI

FastAPI integration lives in `gemstone_py.aio.fastapi`.

```
from fastapi import Depends, FastAPI
from gemstone_py import GemStoneConfig
from gemstone_py.aio import AsyncSession
from gemstone_py.aio.fastapi import session_dependency

app = FastAPI()
get_gemstone = session_dependency(config=GemStoneConfig.from_env())

@app.get("/health/gemstone")
async def gemstone_health(session: AsyncSession = Depends(get_gemstone)):
    return {"result": await session.eval("3 + 4")}
```

Use this when your web stack is async-first. Use the Flask integration when your application is Flask-first.

The runnable repository example is `examples/fastapi/app.py`.

Benchmarks and Build Lanes

This package has both examples and maintained operational lanes. Keep them separate in your head.

Examples are for:

- learning
- translation exercises

- inspection

The benchmark lane is for:

- reproducible measurement
- stored JSON reports
- baseline comparison
- GitHub workflow enforcement

The CLI is:

```
gemstone-benchmarks --entries 500 --search-runs 20
```

Release and Packaging Surface

The public install surface is the `gemstone_py` package.

The package now has:

- PyPI publishing
- TestPyPI rehearsal
- post-release verification against real PyPI
- installed-artifact API contract checks
- optional PyO3 native wheels through `gemstone-py[fast]`

That means you can treat the package as a real distributable unit, not just a working directory with ambition.

Native Fast Path

The default install uses the pure-ctypes GCI backend:

```
python -m pip install gemstone-py
```

The optional native fast path installs the PyO3 extension package:

```
python -m pip install "gemstone-py[fast]"
```

Backend selection happens at import time. For comparisons, set `GEMSTONE_PY_GCI_BACKEND=ctypes` or `GEMSTONE_PY_GCI_BACKEND=native` before starting Python.

The runnable repository example is `examples/native_backend/check_backend.py`.

Recommended Adoption Path

If you are bringing `gemstone-py` into a project, a sensible order is:

1. start with `GemStoneConfig` + `GemStoneSession`
2. move most application work into `session_scope(...)`
3. use `PersistentRoot` first for obvious top-level state
4. introduce `GSCollection` only when indexed queries are justified
5. add `TypedOop[T]`, typed queries, or managed handles when the code benefits

from clearer object identity and lifetime

6. use Flask or FastAPI providers once request lifecycles matter
7. install `gemstone-py[fast]` only after the ctypes path is working
8. add benchmarks and live tests once the system is real

That order keeps the learning curve honest without making the first week feel like a database theology seminar.

Manual Summary

If you remember only five things:

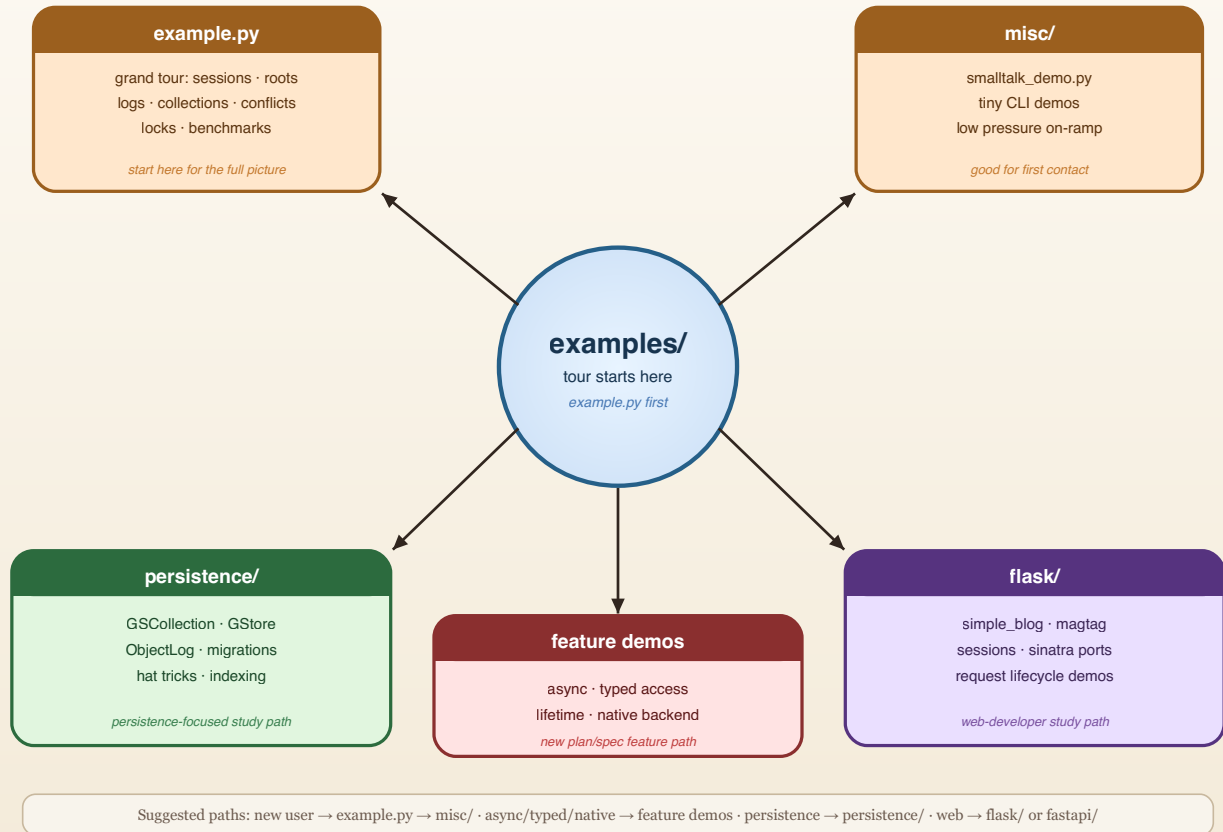
1. Be explicit about transaction policy.
2. Use `PersistentRoot` first unless you need something more specialized.
3. Use async wrappers for async frameworks, not for shared-session threading.
4. Use typed and managed OOP wrappers when raw integers hide too much intent.
5. The examples teach the package; the maintained workflows prove it.

Guide to the Examples

The `examples/` tree is one of the best parts of `gemstone-py`. It is broad enough to teach the package properly, but disciplined enough that the examples still map to the real API and current workflows.

Map of the Examples Directory

Start in the centre, then wander toward the demo that matches your appetite.



How to Use This Guide

This guide is organized around what you are trying to learn:

- first contact with a live GemStone session
- async sessions and FastAPI request integration
- typed OOPs, typed queries, and export-set lifetime handles
- persistence helpers
- indexed collections and query-style access
- concurrency primitives
- Flask integration
- optional native backend selection
- web app translations
- maintained benchmarks versus teaching examples

If you want the shortest path:

1. run `examples/example.py`

2. run `examples/misc/smalltalk_demo.py`
3. run one feature example from `async_features/`, `typed_access/`, or `lifetime/`
4. inspect one persistence example
5. inspect one Flask or FastAPI example

Common Setup

Most live examples expect the same GemStone environment:

```
export GS_LIB=/opt/gemstone/product/lib
export GS_STONE=gs64stone
export GS_USERNAME=DataCurator
export GS_PASSWORD=swordfish
```

Run examples as modules from the repository root so imports resolve exactly as they do in CI:

```
python -m examples.misc.smalltalk_demo
python -m examples.async_features.session_root_and_collection
```

For a source checkout with development tools:

```
python -m pip install -e .[dev]
```

For the optional native backend example:

```
python -m pip install "gemstone-py[fast]"
```

`examples/example.py`: The Grand Tour

If you only run one example, run this one.

Why it matters:

- it exercises the major package surfaces in one file
- it demonstrates live session behaviour instead of abstract promises
- it shows transaction boundaries, cross-session reads, and conflict patterns

It covers:

- `SmalltalkBridge`
- `GemStoneSessionFacade`
- `PersistentRoot`

- `GStore`
- `ObjectLog`
- `RCCounter`
- `RCHash`
- `RCQueue`
- nested transactions
- `CommitConflictError`
- date/time conversions
- object locking
- instance listing

Treat it as the "here is what the package can really do" demo.

Usage:

```
python -m examples.example
```

The core shape is still ordinary session code:

```
from gemstone_py import GemStoneConfig, GemStoneSession, TransactionPolicy
from gemstone_py.session_facade import GemStoneSessionFacade

config = GemStoneConfig.from_env()

with GemStoneSession(
    config=config,
    transaction_policy=TransactionPolicy.COMMIT_ON_SUCCESS,
) as session:
    facade = GemStoneSessionFacade(session)
    facade["ExampleDict"] = {"status": "ok"}
```

examples/misc/

This is the low-pressure on-ramp.

Start here when:

- you want a tiny first success
- you want to confirm the environment is healthy
- you want a bridge demo without reading a hundred moving parts

The star here is `smalltalk_demo.py`, which now runs through the supported CLI surface and shows the Smalltalk bridge without dragging in everything else.

Usage:

```
python -m examples.misc.smalltalk_demo
gemstone-smalltalk-demo
gemstone-examples smalltalk-demo
```

Representative code:

```
from gemstone_py.example_support import MANUAL_POLICY, example_session
from gemstone_py.smalltalk_bridge import SmalltalkBridge

with example_session(transaction_policy=MANUAL_POLICY) as session:
    smalltalk = SmalltalkBridge(session)
    settings = smalltalk.StringKeyValueDictionary.new()
    settings["status"] = "ok"
    print(settings["status"])
```

examples/async_features/

This directory demonstrates the async API added for modern Python application stacks.

`session_root_and_collection.py` shows:

- `AsyncSession.connect(...)`
- `async with session.transaction()`
- `AsyncPersistentRoot`
- `AsyncManagedOop`
- `AsyncGSCollection`

Run it when the normal sync login works and you want to confirm that GCI work is being routed through the async facade correctly.

Usage:

```
python -m examples.async_features.session_root_and_collection
```

Session and root access:

```
from gemstone_py.aio import AsyncPersistentRoot, AsyncSession

async with AsyncSession.connect(config=config) as session:
    async with session.transaction():
        root = AsyncPersistentRoot(session)
        await root.set("AsyncExample", {"status": "async"})

    saved = await session.root().get("AsyncExample")
```

Async managed OOPs:

```
ref = await session.execute_managed("OrderedCollection new")
try:
    first = await session.new_string("from async")
    await ref.send("add:", first)
    print(await ref.print_string())
finally:
    await ref.close()
```

Async indexed collection access:

```
from gemstone_py.aio import AsyncGSCollection

collection = AsyncGSCollection("AsyncFeaturePeople", config=config)
await collection.bulk_insert([
    {"@name": "Ada", "@status": "active"},
    {"@name": "Grace", "@status": "inactive"},
])
await collection.add_index("@status")
active = await collection.search("@status", "eq", "active")
collection.close()
```

Use a generated or test-specific collection name in real scripts, and call `AsyncGSCollection.drop(...)` during cleanup.

examples/typed_access/

This directory covers the static typing stream from the plan.

Use it when you want to understand:

- `@gemstone_class(...)`
- `TypedOop[T]`
- `typed_oop(...)`
- `typed GSCollection.query(Protocol)` predicates

`simple_blog_queries.py` contains importable helper functions for the blog data shape.

`typed_oops_and_queries.py` is the runnable live example.

Usage:

```
python -m examples.typed_access.typed_oops_and_queries
```

Typed OOPs:

```

from typing import Protocol
from gemstone_py import GemStoneSession, TransactionPolicy, gemstone_class

@gemstone_class("Date")
class GemStoneDate(Protocol):
    @property
    def printString(self) -> str:
        ...

with GemStoneSession(
    config=config,
    transaction_policy=TransactionPolicy.ABORT_ON_EXIT,
) as session:
    today = session.execute_typed("Date today", GemStoneDate) # type: ignore[type-abstract]
    print(today.gemstone_class_name)
    print(today.proxy().printString)

```

Typed `GSCollection` queries:

```

from typing import Protocol
from gemstone_py.gsquery import GSCollection

class BlogPostRecord(Protocol):
    title: str
    status: str

posts = GSCollection("SimplePosts", config=config)
published = posts.query(BlogPostRecord).where(
    lambda post: post.status == "published"
).all()

for post in published:
    print(post.title)

```

The lambda records a GemStone ivar path. It is not a Python-side row filter.

examples/lifetime/

`managed_oop_handles.py` is the focused example for the OOP lifetime model.

It demonstrates:

- `execute_managed(...)` for automatic export-set retention
- `ManagedOop.close()` for explicit cleanup
- `session.handle(raw_oop)` for scoped retention when you already have an OOP

Read this before holding raw OOP integers across operations that might trigger GemStone-side garbage collection.

Usage:

```
python -m examples.lifetime.managed_oop_handles
```

Managed handle:

```
collection = session.execute_managed("OrderedCollection new")
try:
    collection.send("add:", session.new_string("managed"))
    session.eval("System startGcAndCommit")
    print(collection.print_string())
finally:
    collection.close()
```

Scoped raw-OOP handle:

```
raw_oop = session.execute_oop("OrderedCollection new")
with session.handle(raw_oop) as handle:
    handle.send("add:", session.new_string("scoped"))
    print(handle.send("printString"))
```

Use `execute_managed(...)` for object references you want Python to retain automatically. Use `session.handle(...)` when an existing raw OOP only needs to be protected inside a known block.

examples/native_backend/

`check_backend.py` prints the selected low-level GCI backend and whether the optional native extension is importable.

Use it after:

```
python -m pip install "gemstone-py[fast]"
```

It is also useful when comparing benchmark reports with `GEMSTONE_PY_GCI_BACKEND=ctypes` and `GEMSTONE_PY_GCI_BACKEND=native`.

Usage:

```
python -m examples.native_backend.check_backend
GEMSTONE_PY_GCI_BACKEND=ctypes python -m examples.native_backend.check_backend
GEMSTONE_PY_GCI_BACKEND=native python -m examples.native_backend.check_backend
```

The code is deliberately tiny:

```
from gemstone_py import _gci
from gemstone_py.native import native_fast_path_available

print(_gci.IMPLEMENTATION)
print(native_fast_path_available())
```

Backend selection happens when Python imports `gemstone_py._gci`, so set `GEMSTONE_PY_GCI_BACKEND` before starting Python.

examples/persistence/

This is where the package stops being a client library and starts feeling like a practical application toolkit.

Major themes in this tree:

- indexed collections
- stores
- data migration
- persistent data structures
- translation of old MagLev-era ideas into plain GemStone use

Indexing

Files around `persistence/indexing/` show how to:

- create a dataset
- store rows in GemStone
- build indexes
- search efficiently

This is the right cluster to study when `PersistentRoot` starts to feel too coarse-grained for the shape of your data.

Usage:

```
python -m examples.persistence.indexing.create_random_people
python -m examples.persistence.indexing.search_random_people
python -m examples.persistence.indexing.index_example
```

The useful API shape:

```
from gemstone_py.gsquery import GSCollection

people = GSCollection("People", config=config)
people.bulk_insert([
```

```

    {"@name": "Ada", "@age": 36},
    {"@name": "Grace", "@age": 42},
  ])
  people.add_index_for_class("@age", "SmallInt")
  matches = people.search("@age", "gte", 40)

```

GStore

The `persistence/gstore/` example contrasts GemStone-backed storage with an in-memory dict baseline. That makes it useful for both teaching and performance intuition:

- you see the API shape
- you see the cost of persistence
- you do not need mythology to explain the result

Usage:

```
python -m examples.persistence.gstore.main
```

Typical store-shaped code:

```

from gemstone_py.gstore import GStore

store = GStore("inventory.db", config=config)
store["sku:hat"] = {"name": "Hat", "stock": 8}
print(store["sku:hat"])

```

Hat Trick

The `hat_trick/` example is the kind of thing every useful repository-backed toolkit should have: a weird little demo that is memorable enough to teach the real abstraction.

Here the abstraction is queue-backed shared state:

- create a hat backed by `RCQueue`
- inspect the contents
- understand the relation between a playful example and a real concurrency primitive

The queue is not a toy. The hat is merely wearing formal attire.

Usage:

```

python -m examples.persistence.hat_trick.create_hat
python -m examples.persistence.hat_trick.add_rabbit_to_hat
python -m examples.persistence.hat_trick.show_hat_contents

```

The pattern behind the example is reduced-conflict queue-backed state:

```
from gemstone_py.concurrency import RCQueue
from gemstone_py.persistent_root import PersistentRoot

root = PersistentRoot(session)
root["WorkQueue"] = RCQueue(session)
root["WorkQueue"].push("first job")
```

Migrations

The migrations examples are valuable because they show the package in an honest "old data must become new data" mode.

Use these when you want patterns for:

- versioned domain objects
- chunked migration
- retry loops
- explicit transactional migration steps

Usage:

```
python -m examples.persistence.migrations.write_posts
python -m examples.persistence.migrations.migrate
python -m examples.persistence.migrations.migrate_by_chunks
```

The key habit is to keep migration units explicit:

```
from gemstone_py import session_scope

with session_scope(config=config) as session:
    # read old shape
    # write new shape
    # commit happens only if the block succeeds
    ...
```

examples/flask/

This directory is where the request/session layer stops being theory.

Important sub-examples:

- `simple_blog/`
- `magtag/`
- `sessions/`

- `sinatra_port/`

`simple_blog`

Good first Flask example because:

- the domain is small
- the persistence is recognizable
- the routes are ordinary enough to reason about quickly

Study this when you want:

- request session handling
- a simple persistent model
- proof that the package fits normal Flask structure

Usage:

```
python -m examples.flask.simple_blog.blog_app
```

Request-scoped work should use the installed session integration:

```
from flask import Flask
from gemstone_py import GemStoneConfig, install_flask_request_session

app = Flask(__name__)
install_flask_request_session(
    app,
    config=GemStoneConfig.from_env(),
    pool_size=4,
)
```

`magtag`

This is a larger demonstration of the same core idea:

- Flask app
- GemStone-backed data
- `GSCollection` in a realistic setting

If `simple_blog` is the friendly coffee, `magtag` is the "all right, show me a real screen and some real workflows" example.

Usage:

```
python -m examples.flask.magtag.magtag_app
```

The model layer uses `GSCollection` for lookup-heavy records:

```
from gemstone_py.gsquery import GSCollection

users = GSCollection("MagTagUsers")
users.add_index_for_class("@name", "String")
matches = users.search("@name", "eq1", "pbm0")
```

sessions

These examples focus on request/session lifecycle concerns. They are useful when you care more about integration behaviour than about application domain logic.

Usage:

```
python -m examples.flask.sessions.msessions
```

Provider health can be inspected without reaching into private Flask state:

```
from gemstone_py import flask_request_session_provider_snapshot

snapshot = flask_request_session_provider_snapshot(app)
if snapshot is not None:
    print(snapshot.available, snapshot.in_use, snapshot.created)
```

examples/fastapi/

This is the smallest async web example. It uses `session_dependency(...)` to open one `AsyncSession` per request and commit or abort with the request outcome.

Usage:

```
python -m pip install fastapi uvicorn
uvicorn examples.fastapi.app:app --reload
```

Endpoint shape:

```
from fastapi import Depends, FastAPI
from gemstone_py import GemStoneConfig
from gemstone_py.aio import AsyncSession
from gemstone_py.aio.fastapi import session_dependency
```

```

app = FastAPI()
get_gemstone_session = session_dependency(config=GemStoneConfig.from_env())

@app.get("/health/gemstone")
async def gemstone_health(session: AsyncSession = Depends(get_gemstone_session)):
    return {"result": await session.eval("3 + 4")}

```

examples/django/ **and** examples/webstack/

These matter for two reasons:

1. they prove the package is not locked into one tiny application style
2. they surface compatibility issues that smaller demos never trigger

The `webstack` examples are also useful for release verification because they have enough app surface to catch missing imports, route regressions, and other "this only broke in CI" problems.

Usage:

```

cd examples/django/myapp
python manage.py migrate
python manage.py runserver

```

```
python -m examples.webstack.magtag_app
```

Examples vs Maintained Lanes

This distinction matters.

The examples are for learning and exploration.

The maintained lanes are:

- `./scripts/run_ci_checks.sh`
- `./scripts/run_live_checks.sh`
- `gemstone-benchmarks`
- the GitHub workflows under `.github/workflows/`

Do not confuse:

- "I ran a charming example"
- with
- "the package is verified against its supported workflows"

You need both. They are not interchangeable.

Suggested Study Paths

Path 1: New user

1. `examples/example.py`
2. `examples/misc/smalltalk_demo.py`
3. this guide again, now with less fear

Path 2: Persistence-focused user

1. `examples/example.py`
2. `examples/persistence/indexing/*`
3. `examples/persistence/gstore/main.py`
4. `examples/persistence/migrations/*`

Path 3: Async or FastAPI user

1. `examples/async_features/session_root_and_collection.py`
2. `examples/fastapi/app.py`
3. `tests/test_live_async_integration.py`

Path 4: Typed API user

1. `examples/typed_access/typed_oops_and_queries.py`
2. `examples/typed_access/simple_blog_queries.py`
3. `tests/test_typed_oop_lifetime.py`
4. `tests/test_gsquery.py`

Path 5: Web developer

1. `examples/flask/simple_blog/*`
2. `examples/flask/sessions/*`
3. `examples/flask/magtag/*`
4. `examples/fastapi/app.py`
5. `examples/webstack/*`

Path 6: Concurrency-curious person

1. `examples/example.py`
2. hat trick queue demo
3. live tests around contention and retries

Path 7: Packaging and performance user

1. `examples/native_backend/check_backend.py`
2. `gemstone-benchmarks --suite gci --entries 1000000`
3. `./scripts/run_native_checks.sh`

Practical Advice

- run the examples from a configured environment, not from a shell that "mostly remembers" the right variables
- do not start with the biggest web example if you still do not know how `TransactionPolicy` works
- read the maintained benchmark docs before treating example performance output as policy
- keep the examples open next to the user manual; the two reinforce each other

Closing Thought

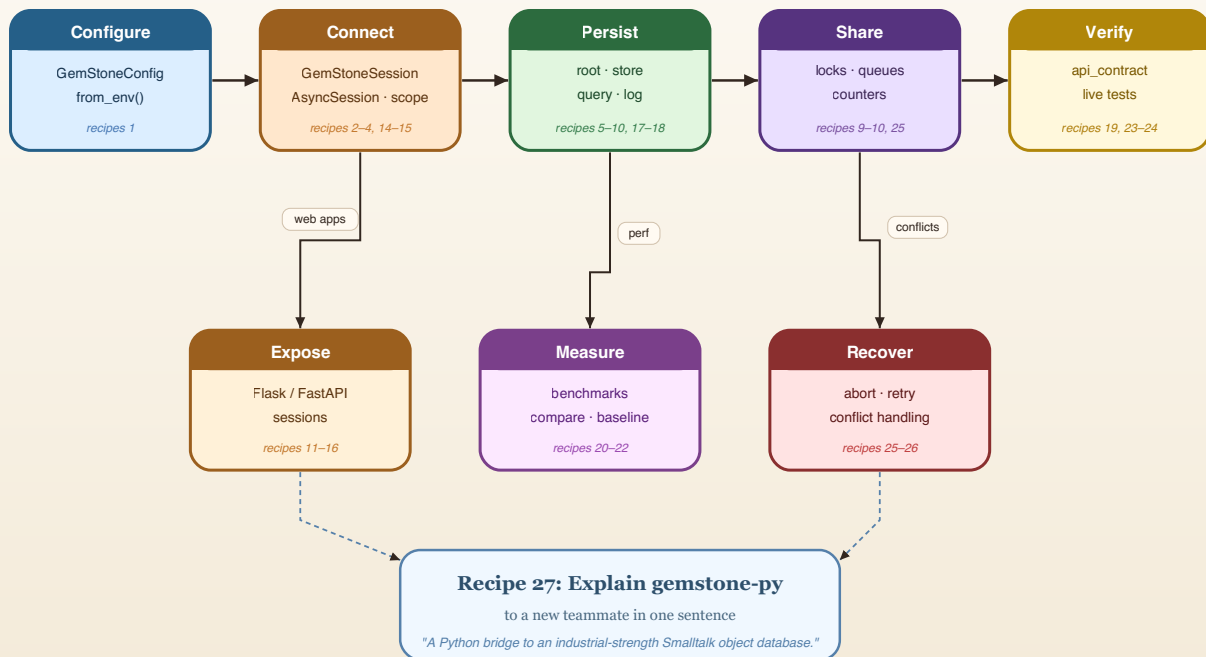
The examples directory is not filler. It is one of the package's strongest teaching tools, and it is worth reading like source documentation instead of like a pile of promotional demos.

Cookbook

This cookbook is a collection of direct recipes. Each one is deliberately short enough to copy, adapt, and keep moving.

Cookbook Recipe Flow

From config to production: how the 27 cookbook recipes connect.



Reading order:

Follow the top row left to right for the main happy path, then drop down as your use case demands.

Recipe 1: Open a Session and Evaluate Smalltalk

```
from gemstone_py import GemStoneConfig, GemStoneSession

config = GemStoneConfig.from_env()

with GemStoneSession(config=config) as session:
    print(session.eval("3 factorial"))
```

Use this when:

- you want the smallest live sanity check
- you need to inspect a repository value quickly

Recipe 2: Commit a Write Safely

```
from gemstone_py import GemStoneConfig, TransactionPolicy, GemStoneSession
from gemstone_py.persistent_root import PersistentRoot

config = GemStoneConfig.from_env()

with GemStoneSession(
```

```

    config=config,
    transaction_policy=TransactionPolicy.COMMIT_ON_SUCCESS,
) as session:
    root = PersistentRoot(session)
    root["CookbookExample"] = {"answer": 42}

```

Why this recipe exists:

- it avoids the classic "the write seemed to work but vanished" mistake

Recipe 3: Read From PersistentRoot

```

from gemstone_py import GemStoneSession
from gemstone_py.persistent_root import PersistentRoot

with GemStoneSession(config=config) as session:
    root = PersistentRoot(session)
    print(root["CookbookExample"])

```

Recipe 4: Use session_scope(...)

```

from gemstone_py import GemStoneConfig, session_scope
from gemstone_py.persistent_root import PersistentRoot

config = GemStoneConfig.from_env()

with session_scope(config=config) as session:
    root = PersistentRoot(session)
    root["ScopedWork"] = {"kind": "unit-of-work"}

```

This is the recipe to prefer in application code.

Recipe 5: Create a Named GSCollection

```

from gemstone_py.gsquery import GSCollection

people = GSCollection("CookbookPeople", config=config)
people.insert({"@name": "Ada", "@city": "London"})
people.insert({"@name": "Grace", "@city": "New York"})
people.add_index("@city")

```

Recipe 6: Search an Indexed Collection

```
matches = people.search("@city", "eq1", "London")
for row in matches:
    print(row)
```

Use `GSCollection` when you need repeated search, not when you merely enjoy the idea of repeated search.

Recipe 7: Keep a Store-Like Dataset in GStore

```
from gemstone_py.gstore import GStore

store = GStore("cookbook.db", config=config)
store["sku:hat"] = {"name": "Hat", "stock": 8}
store["sku:cape"] = {"name": "Cape", "stock": 3}
print(store["sku:hat"])
```

Recipe 8: Append a Persistent Event Log Entry

```
from gemstone_py.objectlog import ObjectLog

log = ObjectLog(config=config)
log.info("inventory_adjusted", {"sku": "hat", "delta": -1})
```

Later:

```
for entry in log.entries():
    print(entry)
```

Recipe 9: Share a Counter Between Sessions

```
from gemstone_py.concurrency import RCCounter

with session_scope(config=config) as session:
    counter = RCCounter(session)
    counter += 1
```

Use shared counters when the number truly lives in GemStone, not when a local integer would do and you are simply feeling theatrical.

Recipe 10: Share a Queue

```
from gemstone_py.concurrency import RCQueue
from gemstone_py.persistent_root import PersistentRoot

with session_scope(config=config) as session:
    root = PersistentRoot(session)
    root["WorkQueue"] = RCQueue(session)
```

Recipe 11: Install Flask Request Sessions

```
from flask import Flask
from gemstone_py import GemStoneConfig, install_flask_request_session

app = Flask(__name__)

install_flask_request_session(
    app,
    config=GemStoneConfig.from_env(),
    pool_size=4,
)
```

This is the default production-friendly shape.

Recipe 12: Use a Thread-Local Provider for Simpler Hosting

```
install_flask_request_session(
    app,
    config=GemStoneConfig.from_env(),
    thread_local=True,
)
```

Good when the server model is simple and you want one session per thread.

Recipe 13: Inspect Provider Health

```
from gemstone_py import flask_request_session_provider_snapshot

@app.get("/health/gemstone")
```

```
def gemstone_health():  
    return flask_request_session_provider_snapshot()
```

Operational visibility is better than optimism.

Recipe 14: Open an Async Session

```
from gemstone_py import GemStoneConfig  
from gemstone_py.aio import AsyncSession  
  
config = GemStoneConfig.from_env()  
  
async with AsyncSession.connect(config=config) as session:  
    print(await session.eval("3 + 4"))
```

Use this when the surrounding application is already async.

Recipe 15: Use an Async Transaction

```
async with AsyncSession.connect(config=config) as session:  
    async with session.transaction():  
        root = session.root()  
        await root.set("AsyncCookbook", {"status": "ok"})
```

The transaction context commits on success and aborts on exception.

Recipe 16: Add FastAPI Dependency Injection

```
from fastapi import Depends, FastAPI  
from gemstone_py import GemStoneConfig  
from gemstone_py.aio import AsyncSession  
from gemstone_py.aio.fastapi import session_dependency  
  
app = FastAPI()  
get_gemstone = session_dependency(config=GemStoneConfig.from_env())  
  
@app.get("/health/gemstone")  
async def gemstone_health(session: AsyncSession = Depends(get_gemstone)):  
    return {"result": await session.eval("3 + 4")}
```

Recipe 17: Query With a Typed Protocol

```
from typing import Protocol
from gemstone_py.gsquery import GSCollection

class PostRecord(Protocol):
    title: str
    status: str

posts = GSCollection("SimplePosts", config=config).query(PostRecord)
for post in posts.where(lambda row: row.status == "published").all():
    print(post.title)
```

The lambda records an indexed GemStone path. It is not called for every row in Python.

Recipe 18: Keep an OOP Alive Explicitly

```
with GemStoneSession(config=config) as session:
    ref = session.execute_managed("OrderedCollection new")
    ref.send("add:", session.new_string("kept"))
    print(ref.print_string())
    ref.close()
```

For a raw OOP:

```
with session.handle(raw_oop) as handle:
    print(handle.send("printString"))
```

Recipe 19: Check the Native Backend

```
python -m pip install "gemstone-py[fast]"
python -m examples.native_backend.check_backend
```

Force a backend before Python starts:

```
GEMSTONE_PY_GCI_BACKEND=ctypes python -m examples.native_backend.check_backend
GEMSTONE_PY_GCI_BACKEND=native python -m examples.native_backend.check_backend
```

Recipe 20: Run the Maintained Benchmarks

```
gemstone-benchmarks --entries 500 --search-runs 20
```

Or emit JSON:

```
gemstone-benchmarks --json --output benchmark-report.json
```

Recipe 21: Compare Benchmark Reports

```
gemstone-benchmark-compare old.json new.json --json --output compare.json
```

That turns performance arguments into evidence, which is disappointingly healthy.

Recipe 22: Register a New Accepted Benchmark Baseline

```
gemstone-benchmark-baseline-register \  
  benchmark-report.json \  
  --manifest .github/benchmarks/index.json
```

Recipe 23: Verify the Installed Artifact

```
python -m gemstone_py.api_contract --json
```

This is useful after installation, after release, and after any moment when you feel your package metadata may have become sentient.

Recipe 24: Run the Live Test Lane

```
GS_RUN_LIVE=1 ./scripts/run_live_checks.sh
```

Longer soak run:

```
GS_RUN_LIVE=1 GS_RUN_LIVE_SOAK=1 ./scripts/run_live_checks.sh
```

Recipe 25: Handle Commit Conflicts Without Pretending They Are Rare

When multiple sessions modify overlapping state, conflicts are normal. The right pattern:

```
from gemstone_py import GemStoneConfig, GemStoneSession, TransactionPolicy
from gemstone_py.concurrency import CommitConflictError

config = GemStoneConfig.from_env()

for attempt in range(3):
    with GemStoneSession(config=config,
                          transaction_policy=TransactionPolicy.MANUAL) as session:
        try:
            # reload data each attempt — previous read is stale after abort
            root = PersistentRoot(session)
            root["counter"] = (root.get("counter") or 0) + 1
            session.commit()
            break
        except CommitConflictError:
            session.abort()
            if attempt == 2:
                raise
```

Rules:

- keep the write unit small
- reload data after every abort — the previous read is stale
- bound the retry count — do not loop forever
- log conflicts — frequent conflicts signal a design smell, not bad luck

Recipe 26: Learn a Queue With a Hat

The hat trick example is memorable because it teaches a real primitive through a slightly ridiculous scenario. You should keep more examples like that in your own codebase than you probably do.

Recipe 27: Explain gemstone-py to a New Teammate

Use this sentence:

"It is a Python package that talks directly to GemStone Smalltalk, keeps transactions explicit, gives us persistence helpers instead of a fake ORM, supports async, typed access, managed OOP lifetimes, and optional native wheels, and already has real CI, release, benchmark, and live verification lanes."

That sentence has rescued several meetings already. At minimum, it should rescue yours.

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